

Driverless Net 3.2 Tbps (8×448 Gbps) PAM4 Transmission at 500m Using a Chirped Differential MZM DR8 PIC on TFLN

Charles St-Arnault⁽¹⁾, Derek Kita⁽²⁾, Ross Dickson⁽³⁾, Bruce Beggs⁽³⁾, Naim Ben-Hamida⁽³⁾, Christian Reimer⁽²⁾, and David V. Plant⁽¹⁾

⁽¹⁾ Dept. of Electrical and Computer Engineering, McGill University, Montreal, QC, Canada, charles.st-arnault@mail.mcgill.ca

⁽²⁾ HyperLight Corporation, Cambridge, MA, USA

⁽³⁾ Ciena Corporation, Ottawa, ON, Canada

Abstract We report a novel ultra-compact chirped differential electrode layout on TFLN reducing $V_{\pi}L$ by 1.33x on a DR8 PIC enabling driverless 3.2 Tbps (8×448 Gbps) PAM4 with compliant TDECQ under the HD-FEC. ©2026 The Author(s)

Introduction

The IEEE 802.3 standard has experienced recent unprecedented growth. This is accompanied by a technically rich evolution in transceiver architecture and transmission speeds. In the past 5 years alone, more transmission speed standards were introduced than in the preceding 35 years [1]. This evolution has come as a direct response for the exponentially growing needs of AI and machine learning data centers. Back in 2024, demonstrations for intensity-modulated direct-detection (IM/DD) 400G/lane systems were limited to the use of PAM8 with increased digital signal processing (DSP) and FEC complexity [2]. Vendors are now demonstrating 400G/lane DSP engines using PAM4 [3–5]. This rapid progress must be followed by rapid innovation to ensure that key data rates and power consumption targets are met. Operating without the use of RF drivers could provide power consumption savings of roughly 10-15% [6–8]. Driverless operation imposes stringent requirements on the driving voltage requirements of the electro-optic modulator to ensure extinction ratio targets are met. As such, a modulator architecture supporting low drive requirements (sub-1V) is required.

Thin-film lithium niobate has emerged as a scalable material platform that can offer the bandwidths and low voltage drive requirements for the next generation of data center interconnects (DCIs). Still, very few driverless demonstrations on TFLN have been reported to date in IM/DD systems [9]. In addition, differential electrode configurations on TFLN require careful design due to lack of native differential compatibility [10, 11]. In this work, we introduce a novel chirped differential electrode on TFLN that increases the strength of the electric field around one of the waveguides providing a 1.33x reduction in $V_{\pi}L$ at the cost of adding chirp. Using this modulator architecture in a DR8 configuration, we demonstrate driverless

transmission of 8×450 Gbps providing a net aggregate rate of 3.2 Tbps with compliant TDECQ and BER values.

Differential signaling on TFLN

Thin-film lithium niobate (TFLN) is a single-domain crystal material where the ferroelectric domains are aligned in the same direction during production [12, 13]. Modulators built on a single domain material need opposite modulating field directions in each Mach-Zehnder modulator (MZM) arm in order to accumulate opposite phase shifts – a requirement for push-pull modulation [14]. This is typically done using a ground-signal-ground (GSG) electrode layout as shown in Fig. 1 (a) on X-cut TFLN. Each MZM arm is equally modulated by an electric field proportional to $1 \times$ voltage amplitude.

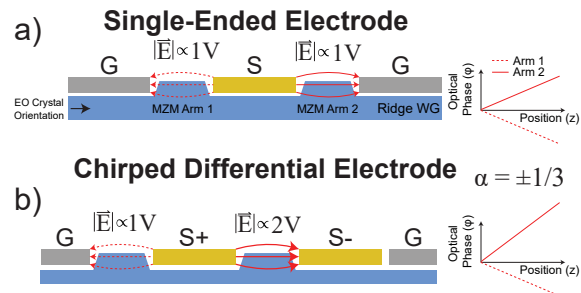


Fig. 1: (a) Standard single-ended electrode layout on X-cut TFLN. The modulating electric field has opposite orientation in each MZM arm to provide opposite phase shift required for push-pull modulation. (b) Chirped differential electrode layout. The lower MZM arm has a modulating electric field with double the amplitude of that in the other arm providing a 1.33x increase in modulation strength.

The modulating electric fields across both waveguides face opposite directions, providing chirp-less push-pull modulation. This electrode configuration is not compatible with differential signaling provided by CMOS DAC outputs. The use of differential signaling for modulation provides a reduction in noise due to common-mode noise

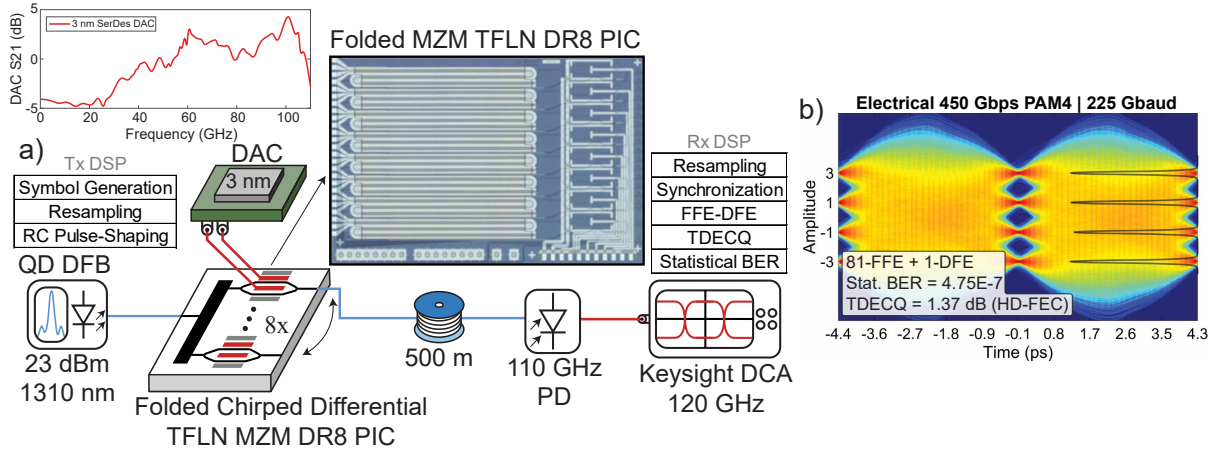


Fig. 2: (a) Experimental setup and DSP stack used in the driverless 3.2 Tbps demonstration. A folded chirped differential MZM DR8 PIC is used. Eye diagrams are captured with a Keysight DCA, with offline processing used to equalize the received waveform and calculate TDECQ and statistical BER. (b) Electrical eye diagram of the 3 nm DAC with 81-FFE and 1-DFE taps with TDECQ of 1.37 dB and statistical BER of 4.75E-7.

and even harmonic rejection at the RF driver side, increases the total RF swing that the driver can deliver, and halves the power consumption of the driver for a fixed total output swing and termination resistance [10]. Furthermore, differential electrode configurations can increase the electric field intensity across the waveguides by a factor of 2 providing reductions in $V_{\pi}L$ product. Notwithstanding these advantages, very few differential electrode MZMs on TFLN have been demonstrated [15, 16], and to date, no demonstrated differential design on TFLN benefits from the increased field amplitude achievable with differential signaling. Here we present a novel chirped differential electrode configuration, as shown in Fig. 1 (b). The lower MZM arm is surrounded by an S+/S- electrode pair providing double the field intensity in that MZM arm compared to the upper MZM arm. As such, modulation strength is increased by a factor of 1.33x compared to the standard single-ended electrode layout shown in (a), whilst providing differential compatibility. This uneven modulation incurs chirp with factor $\alpha = \pm 1/3$ ($\pm$ denotes the chirp sign dependent on bias point) in the optical field causing pulse shortening or broadening in the presence of chromatic dispersion (CD). Pulse compression – occurring when chirp factor and CD signs are opposite – can be advantageous for CD mitigation [17]. Operating in the O-band near the ~ 1310 nm zero-dispersion wavelength of standard SMF minimizes the impact of chirp.

Experimental Setup

The experimental setup and DSP stacks used in the driverless 3.2 Tbps experiment are shown in Fig. 2 (a). We generate 2^{17} symbols by Mersenne Twister and resample them to match the 2-channel 3 nm CMOS SerDes's 225 GSa/s sampling rate. Raised cosine pulse-shaping with a rolloff of 0.3

is used at 112.5 Gbaud. This DAC has a 6-dB bandwidth exceeding 110 GHz, a 7-bit resolution, a differential output voltage of 700-800 mV_{pp} and inductive peaking at 100 GHz as shown in the inset in Fig. 2 (a). The differential electrical PAM signal directly drives the chirped differential TFLN DR8 PIC with 700-800 mV_{pp} differential. A quantum-dot DFB laser centered at 1310 nm with 23 dBm of output power is used at the transmitter. The laser feeds a network of multi-mode interferometers (MMI) and splits the input laser into 8 paths, each feeding one of the 8 MZM on the PIC. Monitor photo diodes used for bias control are added onto the TFLN PIC. A micrograph of the folded chirped differential electrode DR8 PIC is shown as an inset in Fig. 2 (a). We measure one output of the DR8 PIC at a time. The modulated PAM signal travels through 500 meters of single mode optical fiber and is detected by a 110 GHz photo detector. We receive on average 4-5 dBm of optical power, with variations between channels caused by coupling fiber alignment. A Keysight digital-communication analyzer (DCA) with 120 GHz remote electrical head is used to sample the signal at 32 samples per unit-interval (UI). We process the sampled signal offline. We first synchronize the signal and then apply FFE and/or DFE. At 112.5 Gbaud (200G PAM4), we apply only 15 taps of FFE, and at 225 Gbaud (400G PAM4), we apply 81-taps of FFE and 1-tap of DFE. The RF interface on the 3 nm DAC has a total of 12-dB of loss at 110 GHz requiring more FFE taps at high baud rates to compensate for the added RF loss. These RF losses could not be compensated at the transmitter with pre-emphasis due to limited output voltage swing. The equalized eye diagram is then used to compute both TDECQ and statistical BER. A target SER corresponding to the hard-decision (HD) FEC BER (3.5e-3) limit is used for the 400G

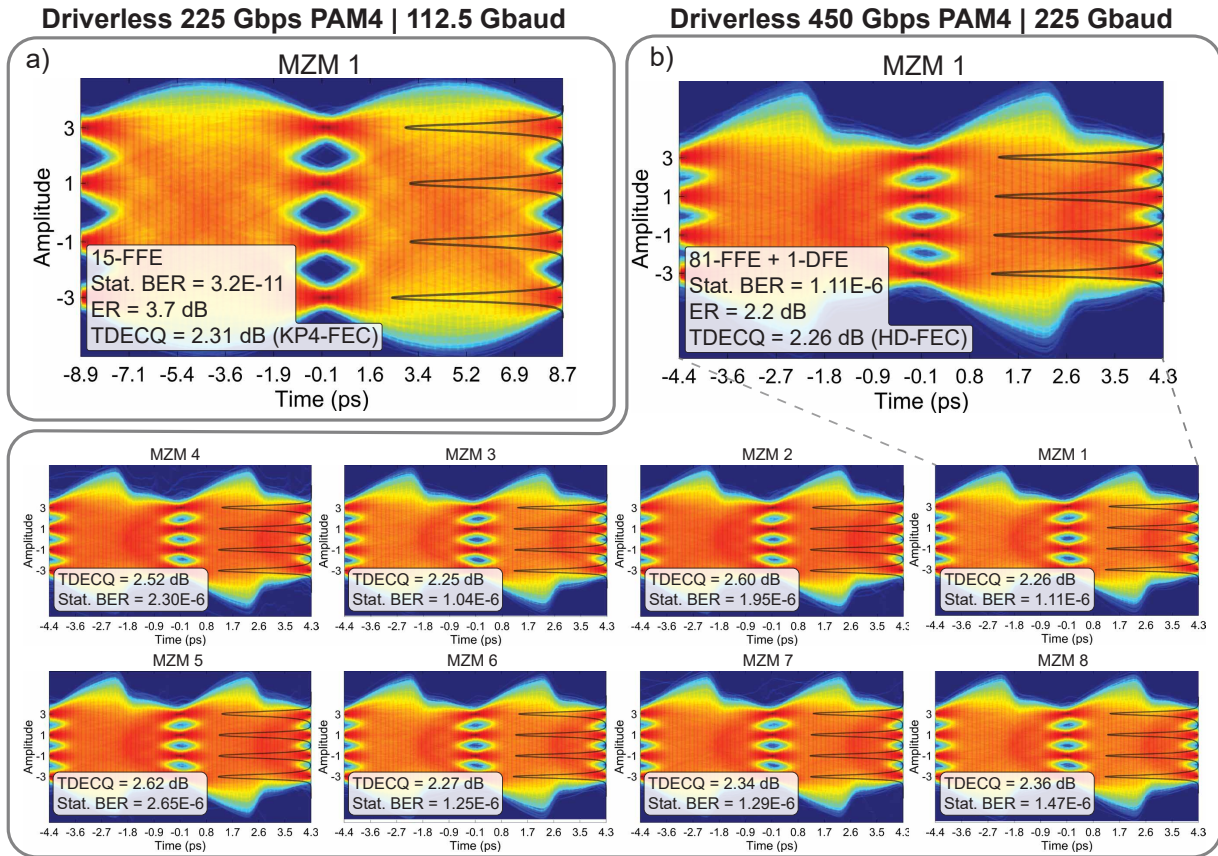


Fig. 3: (a) Optical driverless 225 Gbps PAM4 (112.5 Gbaud) eye diagram. Standard IEEE compliant T-spaced 15-FFE taps are used as the receiver equalizer. A measured pre-equalization extinction ratio (ER) of 3.7 dB is measured, compliant with 200G PMD baseline specifications. (b) Optical driverless 450 Gbps PAM4 (225 Gbaud) eye diagrams for all 8 MZM on the DR8 PIC. Measured TDECQ values across all MZMs remain under 2.62 dB with 81-FFE and 1-DFE equalization. Statistical BERs measured at 0.5 UI remain under $3\text{E-}6$.

measurements, while the standard KP4-FEC target SER is used for the 200G measurements [18]. The electrical eye diagram of the 3 nm DAC is shown in Fig. 2 (c). A statistical BER of $4.75\text{E-}7$ is measured with 81 taps of FFE and 1 DFE tap. A TDECQ of 1.37 dB is measured with the HD-FEC target SER.

Experimental Results and Discussion

The experimental results are shown in Fig. 3. In (a), an optical driverless 225 Gbps PAM4 eye diagram is shown. A standards compliant 15 tap T-spaced FFE is used as the receiver equalizer. The measured extinction ratio (ER) before receiver DSP is 3.7 dB, compliant with 200G/lane specifications [18]. The TDECQ is measured to be 2.31 dB using the KP4 target symbol error rate (SER). At the eye center (0.5 UI), a statistical BER of $3.2\text{E-}11$ is measured. Fig. 3 (b) shows the measured optical eye diagrams across all 8 MZMs on the DR8 PIC for driverless 450 Gbps PAM4 (225 Gbaud). Here, a receiver equalizer consisting of 81 taps of FFE with 1 tap of DFE is used. The high FFE tap count is required to compensate for the DAC RF interface and probing losses that total around 12-dB at 110 GHz. Limited output swing at Nyquist

frequency reduces the measured extinction ratio to 2.2 dB. The measured TDECQ remains below 2.62 dB with an average measured value of 2.40 dB. For these TDECQ calculations, a target SER corresponding to the HD-FEC is used. The average statistical BER is $1.62\text{E-}6$ at the eye opening with all MZMs more than 2 orders of magnitude below the HD-FEC BER limit. The statistical BER remains under $1.5\text{E-}3$ (under HD-FEC BER limit) across a ± 0.08 UI symbol decision instant variation, showcasing resiliency to clock timing jitter. All 8 MZMs perform well under the HD-FEC BER limit demonstrating the first driverless 3.2 Tbps PAM4.

Conclusions

We demonstrated a novel chirped differential electrode architecture on TFLN that provides direct differential signaling compatibility with the CMOS output while reducing the $V_{\pi}L$ product. We implement this electrode configuration with 8 parallel MZMs on a DR8 PIC. We drive all 8 modulators directly from the CMOS DAC without RF drivers and demonstrate net 3.2 Tbps using PAM4.

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